

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

Finding reasoning paths :

- 1 . The anatomical snuff box is a triangular depression on the radial side of the hand , bordered by the tendons of the extensor pollicis longus and brevis . This area is anatomically significant because it contains the radial artery .
- 2 . The radial artery runs through the anatomical snuff box and is superficial , making it vulnerable to injury from sharp objects .
- 3 . A penetrating stab wound in the anatomical snuff box could damage the radial artery , leading to excessive bleeding due to its location and superficial position .

Reasoning Process :

- 1 . The anatomical snuff box is a well-defined area on the radial side of the hand , and it is known to contain the radial artery . This artery is one of the major arteries of the forearm and is responsible for supplying blood to the hand .
- 2 . The radial artery is superficial in the anatomical snuff box , meaning it is closer to the skin and more exposed . This makes it particularly susceptible to injury from sharp objects or penetrating wounds .
- 3 . In the context of the scenario where the teenager received a penetrating stab wound in the anatomical snuff box , the most likely explanation for the excessive bleeding is damage to the radial artery . This is because the artery is directly in line with the wound path and is anatomically positioned to be easily injured .
- 4 . Other structures in the anatomical snuff box , such as tendons and nerves , are deeper and less likely to be involved in a superficial wound . Therefore , the radial artery is the most likely source of the bleeding .

Conclusion :

The radial artery is most likely damaged , leading to the excessive bleeding from the stab wound . This conclusion is based on the anatomical position of the radial artery within the anatomical snuff box and its superficial location , which makes it prone to injury from penetrating wounds .

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<answer>

Radial artery

</answer>

